EVALUATION

I. Choose the best answer

1. Which of the following statements about hydrogen is incorrect ? (NEET - 2016)

   a) Hydrogen ion, H_3O^+ exists freely in solution.

   b) Dihydrogen acts as a reducing agent.

   c) Hydrogen has three isotopes of which tritium is the most common.

   d) Hydrogen never acts as cation in ionic salts.

2. Water gas is

   a) H_2O (g) b) $CO + H_2O$ c) $CO + H_2$ d) $CO + N_2$

3. Which one of the following statements is incorrect with regard to ortho and para dihydrogen ?

   a) They are nuclear spin isomers

   b) Ortho isomer has zero nuclear spin whereas the para isomer has one nuclear spin

   c) The para isomer is favoured at low temperatures

   d) The thermal conductivity of the para isomer is 50% greater than that of the ortho isomer.

4. Ionic hydrides are formed by

   a) halogens b) chalcogens c) inert gases d) group one elements

5. Tritium nucleus contains

   a) 1p + 0 n b) 2 p + 1n c) 1p + 2n d) none of these

6. Non-stoichiometric hydrides are formed by

   a) palladium, vanadium b) carbon, nickel

   c) manganese, lithium d) nitrogen, chlorine

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